Experiment No. 4

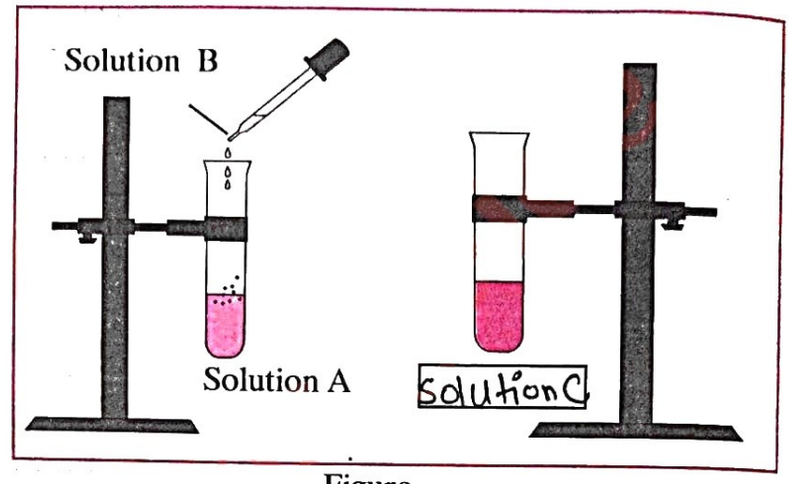
Title : To identify the basic (cation) radicals and acidic (anion) radicals in the given solution.

Apparatus : Test tubes, Stand, Dropper etc.

Chemicals : (A) Copper chloride, (B) Sodium hydroxide, (C) Silver nitrate.

Procedure A :

1. Take the given solution (A) in a test tube.
2. Put 2 to 3 drops of the solution (B) in it with the help of dropper.
3. Observe the colour of the precipitate formed.



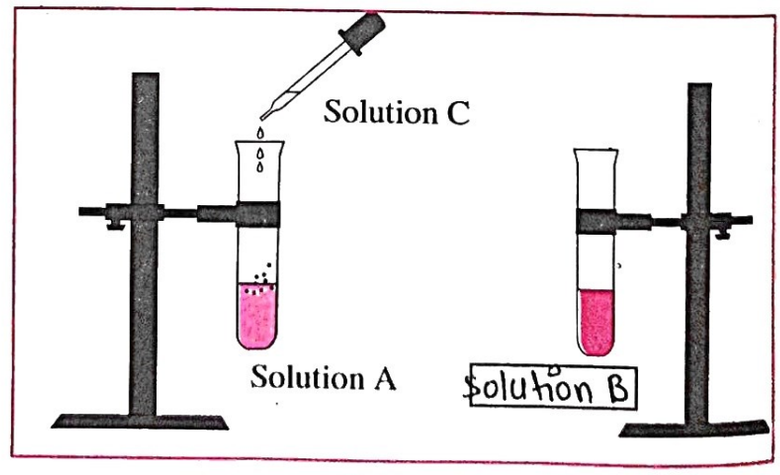
Figure

Observation and Inference :

Procedure	Observation	Inference
Taking the solutions A + B together	The colour of the precipitate is. Blue.	The cation present in the given solution is. cation.

Procedure B :

1. Take the given solution 'A' in a test tube.
2. Put 2 to 3 drops of the solution 'C' in it with the help of dropper.
3. Observe the colour of the precipitate formed.



Figure

Observation and Inference :

Procedure	Observation	Inference
Taking the solution A + C together	The colour of the precipitate is white.	The anion present in the given solution is ....Cl.....

Multiple Choice Questions

1. The precipitate is recognized by the following characteristic.
   A. Floats on the solution B. Settles at the bottomof the solution
   C. Mixes in the solution D. Bubbles are seen in the solution
2. All the acids contain one or more radicals of ...
   A. OH^- B. H^+ C. H^- D. OH^+
3. The composite radicals .....is called hydroxyl radicals.
   A. H^+ B. $(HCO_3)^-$ C. OH^- D. NH_4^+

Questions

1. What is meant by the basicity of an acid?

When we wash cloths soap of turmeric become radish becuse the chemical reaction happen in this and its ~~acid~~ acid is called it basicity of an Acid.

2. What is meant by the acidity of a base?

The number of OH^- ions obtainable by the off ssodation of molecule of an acid is called its acidity.

3. Write down the differences between strong acid and weak acid?

• Strong acid :- an dissolving in water a storg acid dissoted almost completly.
• weak Base :- and dissolving in water a weak bas does not dissociate completly.

4. Complete the following reaction.

$CuCl_2 + NaOH \longrightarrow Cu(OH)_2 + 2NaCl$

$CuCl_2 + AgNO_3 \longrightarrow Cu(NO_3)_2 + 2AgCl$

5. Write down the names of Composite radicals.

The names of composite radicals are
1) Acetylamino
2) methylazo
3) cyclothexane.

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